

Dahil değil

Düzenli D11 ve ifade

$(,), +, -, *$

$(a+b) \rightarrow$ Düzenli ifade

$\downarrow$
 $\begin{bmatrix} a \\ b \end{bmatrix}$

$(a+(b.c))^*$

λ
 a
 $bc bca$
 $aaabc$

$\emptyset$
 a } Düzenli ifadeler.

r_1 ve r_2 düzenli ise

$r_1+r_2, r_1.r_2, r_1^*, (r_1)$ düzenlidir denir.

$\Sigma = \{a, b, c\}$

$(a+bt)^*(c+\emptyset)$ düzenli ifadedir.

$$L(r_1+r_2) = L(r_1) \cup L(r_2)$$

$$L(r_1.r_2) = L(r_1) L(r_2)$$

$$L((r_1)) = L(r_1)$$

$$L(r_1)^* = (L(r_1))^*$$

$\Sigma = \{a, b\}$

$$r = (a+b)^*(a+bb)$$

$$L(r) = \{a, bb, aa, bbb, ba, abbb, aaaa, \dots\}$$

Soru

$\Sigma = \{0, 1\}$

$L = \{w \in \Sigma^* ; w \text{ en az bir çift '0' içersin}\}$

Buna ait düzenli ifadeyi yazınız.

$$r = (0+1)^* 00 (0+1)^*$$

Soru $\Sigma = \{0,1\}$ alfa besinden iřitilen kelimelerde orta arkaaya '0' gelmeyen dille ait dizenli ifadeyi yazınız.

$$r = ((1)^*(0)(1)^*)^*$$

$$r = (1^*01^*)(0+\lambda) + 1^*(0+\lambda)$$

Soru $L((a+bb)^*)$ 5 uzunluğunda kelimeler bulun

abbbb

aaaaa

bbbba

bbabb

Soru $L((ab+b)^*b(a+ab)^*)$ için $w \leq 4$ kelimelerini yazınız?

b

bab

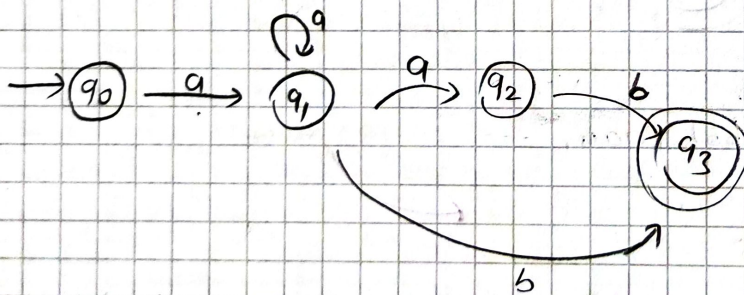
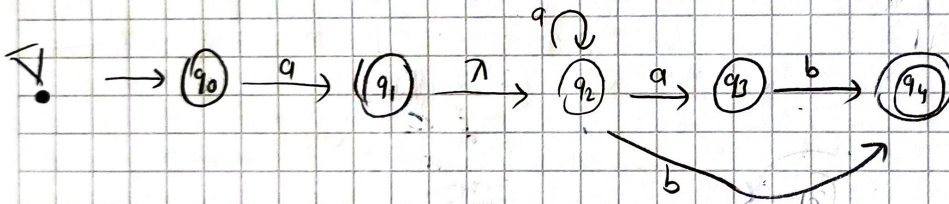
bb

abb

bba

ba

Soru $L(aa^*(ab+b))$ ifadesi için NFA çiziniz?



Soru $L = \{a^n b^m \mid n \geq 3, m \text{ tek}\}$ dille ait dizenli ifade bulunur.

$$r = L(aaa^*b(bb)^*)$$

Soru $L = \{a^n b^m \mid (n+m) \text{ tek}\}$ dizenli ifade bulunur.

$$r = (aa)^*b(bb)^* + (aa)^*a(bb)^*$$

$$L_1 = \{a^n b^m, n \geq 3, m < 4\}$$

$$r = (aaa a^* (\lambda + b + bb + bbb))$$

$$L_2 = \{a^n b^m; n < 4, m \leq 4\} \quad \overline{L_2} \text{ için düzenli ifade yazarız?}$$

λ
 a
 b

aaa bbbb max

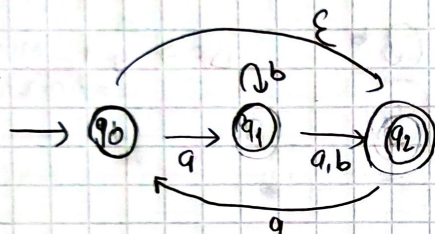
$$\overline{L_2} =$$

$$\overline{L} = \Sigma^* - L$$

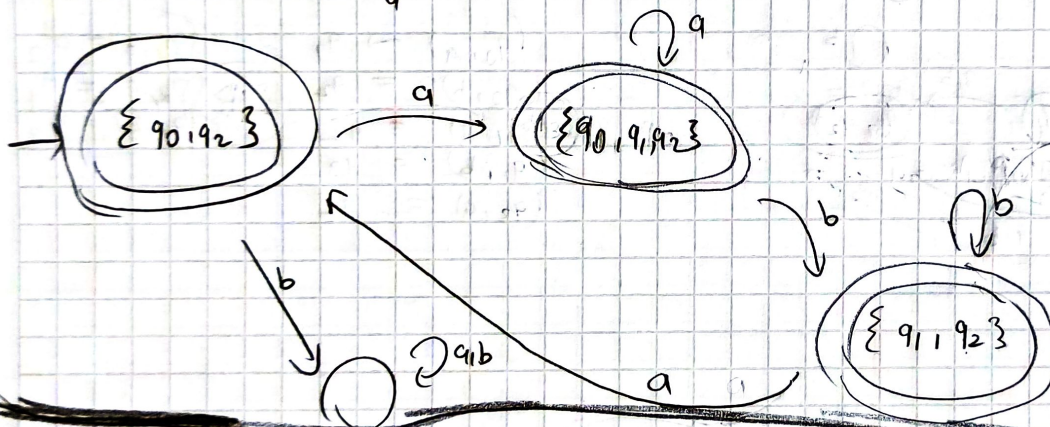
$$\overline{L} = \{a^n b^m; n \geq 4, m \geq 4\}$$

$$r = aaaaa^* bbbbbbb^*$$

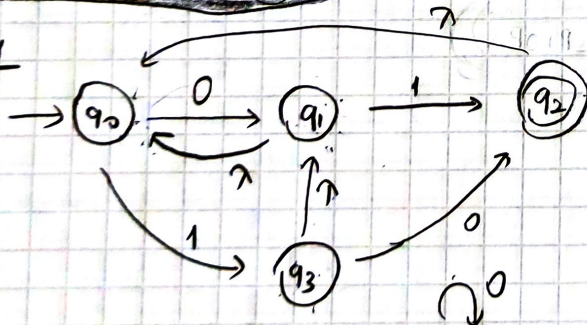
Soru



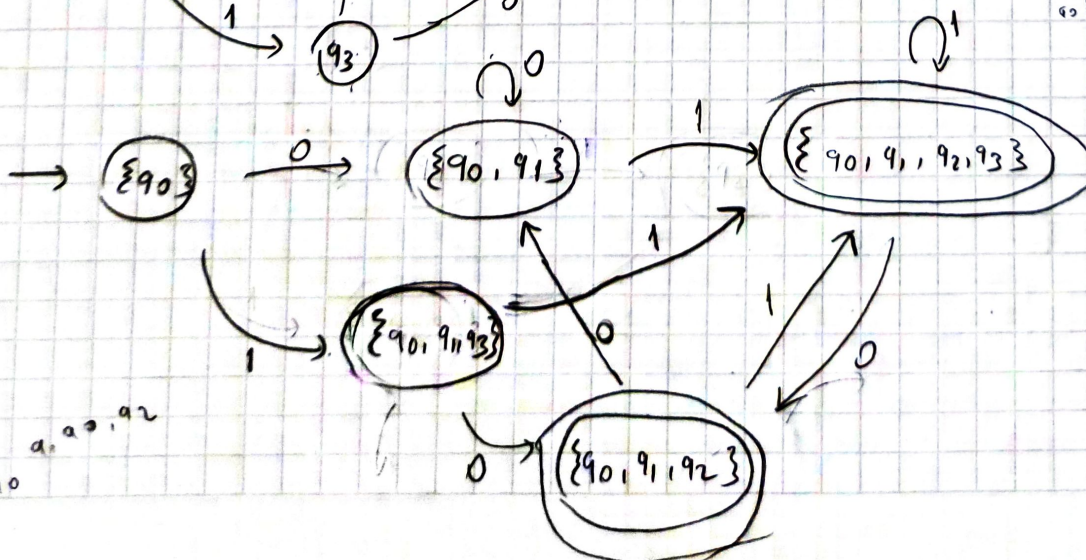
NFA $\rightarrow$ DFA çevir.



Soru



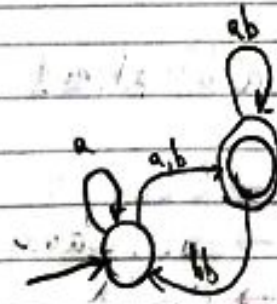
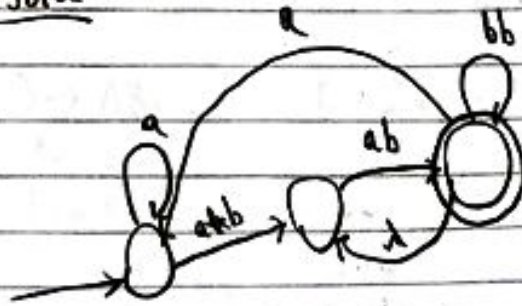
NFA $\rightarrow$ DFA



q_0, q_1, q_2

q_0, q_1, q_2

Soru



$$r = a^*(a+bb)ab((bb)^* + (abbb)^*)^*$$

$$L_r = (ra)^* r$$

Düzenli Gramer

$$G = (V, T, S, P)$$

Sag ve sol lineer olmalı

$$A \rightarrow xB \rightarrow \text{sag lineer}$$

$$A \rightarrow x \quad x \in T^*$$

$$A, B \in V$$

$$A \rightarrow Bx \rightarrow \text{sol lineer}$$

$$A \rightarrow x$$

$$L = \{a^n b^m : n+m \text{ tek}\}$$

$$n \text{ çift } m \text{ tek}$$

$$n \text{ tek } m \text{ çift}$$

$$A \rightarrow B | aC$$

$$B \rightarrow aaB | bD$$

$$C \rightarrow a^2 C | D$$

$$D \rightarrow bbD | \lambda$$

$$L = \{a^n b^m : n \geq 3, m \geq 2\}$$

$$A \rightarrow Bbb$$

$$B \rightarrow Bb | Caaa$$

$$C \rightarrow Ca | \lambda$$

$$L = \{a^n c b^m, n+m \geq 3\} \quad \Sigma = \{a, b, c\}$$

$$S \rightarrow Ac | Bcb | Fcbb | Dcbbb | Ebbb$$

$$A \rightarrow Faal | Aa$$

$$B \rightarrow Faa | Ba$$

$$C \rightarrow Fa | Ca$$

$$F \rightarrow \lambda$$

$$E \rightarrow Eb | Cc | Fc$$

$$F = \{F\}$$

$$L = \{w : n_a(w) - n_b(w) \bmod 3 = 1\}$$

$$S \rightarrow Aaaa | Abbb | Fa | Fb$$

$$F \rightarrow \lambda$$

$$3a - 3b + 1$$

$$3a + 1$$

$$3b + 1$$

$$4a + 1$$

$$(4a + 1) - (3b + 1)$$

$$L = \{w : |n_a(w) - n_b(w)| \text{ tek olsun} \}$$

$$S \rightarrow aaS | bbS | bF | abS | baS | aA | aF | bB$$

$$A \rightarrow bBA | abA | baA | bbF$$

$$B \rightarrow baB | abB | baB | aaF$$

$$F \rightarrow \lambda$$

$$S \rightarrow aaS | bbS | aF | bF | abS | baS$$

$$F \rightarrow \lambda$$

Soru Leftmost - Rightmost

$$S \rightarrow AB$$

$$A \rightarrow aaA$$

$$B \rightarrow Bb$$

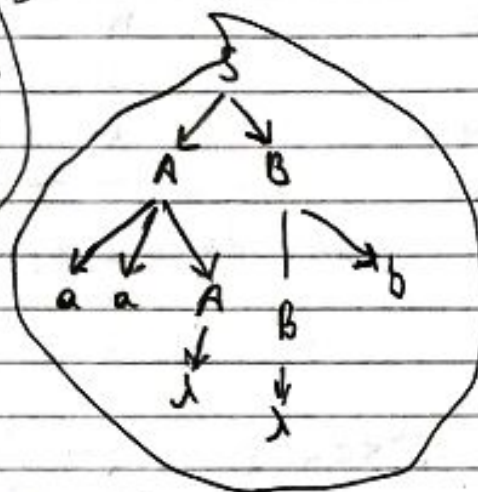
$$B \rightarrow \lambda$$

$$A \rightarrow \lambda$$

$$L = \{a^n b^m : n \geq 0, m \geq 0\}$$

AB
aaAB
aaaaAB
aaaaBb
aaaaab

← leftmost preferred in this word



İçerik Bağımsız Gramer

Tanım Sağ veya sol limer olmayan gramer içerik bağımsızdır

$$S \rightarrow aA$$

$$A \rightarrow Bb$$

$$B \rightarrow Bb \mid \lambda$$

$$L = \{a^n b^n : n \text{ çift}\}$$

$$S \rightarrow \lambda \mid aaAbb$$

$$S \rightarrow aaA \mid \lambda$$

$$A \rightarrow Sbb$$

λ

aaab

aaSbb

aaaaSbbbb

$$L = \{a^n b^n, n = 3k\}$$

$$S \rightarrow a a a A / \lambda$$

$$A \rightarrow S b b b$$

 λ
 a
 aa
 aaa

$$L = \{a^n b^m : n \leq m+3\}$$

 $aaab$
 $aaaaab$

$$S \rightarrow a a A / a A / a b / s b / \lambda$$

 $aaaaab$

$$A \rightarrow a B$$

$$B \rightarrow B b / \lambda / A b$$

 n
 m
 0
 $0, 1, 2, \dots$
 1
 $0, 1, 2, \dots$
 2
 $0, 1, \dots$
 3
 $0, 1, \dots$
 4
 $1, 2, \dots$
 5
 $2, 3, \dots$

$$S \rightarrow \lambda / a b / a a b / a a a b / \dots$$

$$A \rightarrow a A b / B b / \lambda$$

$$B \rightarrow B b / \lambda$$

$$L = \{a^{3m+1}, b^{2n-1}, n^2 < 2m\}$$

 c
 d

$$m \geq 2 \cdot (2 \cdot n // 2 + n \% 2)$$

 n
 m

$$0 \quad m \in (0, \infty)$$

 $a b b$

$$1 \quad m \in (2, \infty)$$

 $aa \quad b b b b b b$

$$2 \quad m \in (4, \infty)$$

$$3 \quad m \in (8, \infty)$$

$$S \rightarrow \lambda / B / a A B B / b$$

$$B \rightarrow b B / b$$

$$A \rightarrow A B B$$

$$A \rightarrow a A B B$$

 $a b b$
 $aa b b$
 $aaa \quad b b b b b b b b$

$$S \rightarrow \lambda / b / b B / a A B B$$

$$B \rightarrow b / b B$$

$$A \rightarrow a B B / \lambda$$

$$A \rightarrow a B B / \lambda$$

$$D \rightarrow A B B$$

 $aaaa$
 $b b b b b b b b b b$

		n	m	
		0	0	

$L = \{a^n b^m, n \neq 2m\}$	$n > 2m$	m	n
	$n < 2m$	0	$(1, \infty)$
$S \rightarrow aA aC bD$		1	$(0, \infty) \neq 1$
$A \rightarrow \lambda Bb$		2	$\neq 4$
$B \rightarrow aaA aC, \lambda$		3	$\neq 6$
$C \rightarrow aC \lambda$		4	$\neq 8$
$D \rightarrow bD \lambda$		5	
		6	
$S \rightarrow aA Bb$		7	
$A \rightarrow Bb$		8	
$B \rightarrow aaA aaA \lambda$		9	
$C \rightarrow$			
$D \rightarrow$			
$\neq ab, aab, aaab, abb, aabb$		m	n
		0	1, 2, ...
		1	2, 3, ...
		2	4, 5, ...
		3	6, 7, ...
		4	8, 9, ...
$aaab$			
$aaaabb$			
	m	n	k
	0	0	0
λ	1	0, 1	
aab	2	0, 1, 2, 3	
$aaabb$	3	0, 1, 2, 3, 4, 5	
$aaaaabb$	4	8	
	5	10	
$aaab$			
$aaaaabb$			

$$L = \{w \in \{a, b\}^* : n_a(w) \neq n_b(w)\}$$

$$S \rightarrow aSb \mid bSa \mid a \mid b \mid ab \mid ba \mid S \mid \epsilon$$

$$S \rightarrow aSb \mid bSa$$

$$S \rightarrow \epsilon$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$L = \{w \in \{a, b\}^* : n_a(w) = n_b(w)\}$$

Işerik Sınırsız Gramer

$$A \Rightarrow x$$

$$A \in V, x \in (V \cup T)^*$$

$$L = \{a^n b^m c^k, k = n + 2m\} \quad n, m, k \geq 0$$

$$n \quad m \quad k \quad \text{SR} \quad S \rightarrow \epsilon \mid aA$$

$$0 \quad 0 \quad 0 \quad A \rightarrow B$$

$$1 \quad 0 \quad 1$$

$$0 \quad 1 \quad 2$$

$$1 \quad 1 \quad 3$$

$$2 \quad 0 \quad 2$$

$$aA b b B c c c c$$

$$S \rightarrow \epsilon \mid aA$$

$$A \rightarrow b b B$$

$$B \rightarrow c c$$

6 sayfa atıldı
diğer pdf'ten
bak

İçerik ~~bağımsız~~ Gramer

$$G = (V, T, S, P)$$

$$A \rightarrow x$$

$$A \in V, x \in (V \cup T)^*$$

Soru

$$L = \{ a^n b^m c^k \mid k = n + 2m \}$$

$$n, m, k \geq 0$$

İçerik bağımsız gramer bulun?

n	m	k
0	0	0
0	1	2
1	0	1
1	1	3
2	1	4
1	2	5

$$S \rightarrow \lambda \mid aA$$

$$A \rightarrow Bbccc \mid aA$$

$$B \rightarrow \lambda$$

$$S \rightarrow \lambda \mid aAc \mid aSc$$

$$A \rightarrow Sbccc \mid \lambda$$

$$S \rightarrow aSc$$

$$aaSc$$

$$aaaAc$$

$$aaaSbcccc$$

$$aaa\lambda bcccc$$

Soru

$$L = \{ a^n b^m c^k \mid k = n + m \}$$

$$n, m, k \geq 0$$

İçerik bağımsız gramer bulun

$$S \rightarrow aAbc$$

$$A \rightarrow aA \mid \lambda$$

$$S \rightarrow aA$$

$$A \rightarrow bB$$

$$B \rightarrow bB \mid c$$

Belirsizlik

Tanım: Aynı kelimeye farklı iki yoldan gidiliyorsa gramer

$S \rightarrow SS \mid Sa \mid a$ belirsizdir denir.

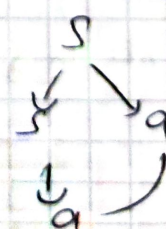
$$S \rightarrow SS$$

$$\rightarrow aa$$

$$S \rightarrow Sa$$

$$\rightarrow aa$$

belirsizlik



Simple Grammar

$$G = (V, T, S, P)$$

$$A \rightarrow ax$$

$$A \in V, a \in T, x \in V^* \quad ! \text{ iki tane terminal sembolü olamaz.}$$

$$S \rightarrow aS \mid bSS \mid c$$

Soru

$L(aaa^*b + ab^*)$ için simple gramer bulun.

Kelimeler

abbb....

aa**a^{*}...**b

$$S \rightarrow aA \mid aB$$

$$A \rightarrow bA \mid \lambda$$

$$B \rightarrow aC$$

$$C \rightarrow ab \mid aC \quad \text{veya} \quad C \rightarrow b \mid aD$$

$$D \rightarrow aD \mid b$$

Soru $\{a^n b^n : n \geq 2\}$ S-grammer?

$$\begin{aligned} S &\rightarrow aA \\ A &\rightarrow aB \\ B &\rightarrow aB \mid bC \\ C &\rightarrow bD \\ D &\rightarrow bD \mid \lambda \end{aligned}$$

$$\begin{aligned} S &\rightarrow @A \\ A &\rightarrow B@ \\ B &\rightarrow aC \\ C &\rightarrow D@ \\ D &\rightarrow aC \mid \lambda \end{aligned}$$

Terminal sembolü solda
Olmali, sağda olmamalı

$$\begin{aligned} S &\rightarrow aAB \\ A &\rightarrow aAB \mid aC \\ B &\rightarrow b \\ C &\rightarrow b \end{aligned}$$

Soru

$L = \{a^n b^{n+1} : n \geq 1\}$ için S-grammer yapınız?

abb
aabbb

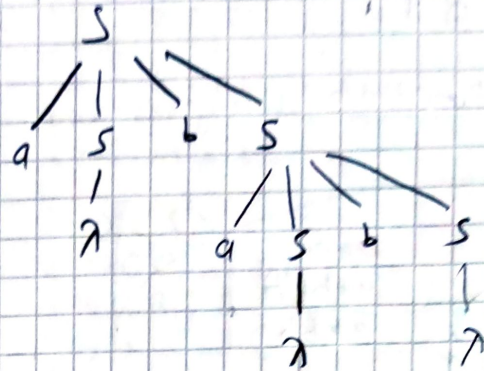
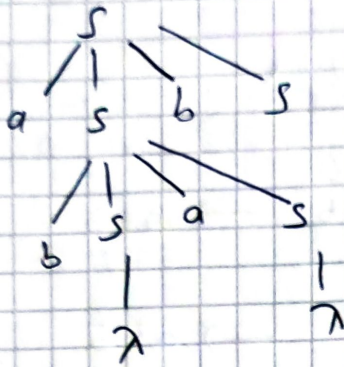
?

$$\begin{aligned} S &\rightarrow aAB \\ A &\rightarrow aA \mid aD \\ B &\rightarrow bC \\ C &\rightarrow bC \mid bD \\ D &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} S &\rightarrow aAB \\ A &\rightarrow aAB \\ B &\rightarrow b \\ A &\rightarrow b \end{aligned}$$

$$S \rightarrow aSbS \mid bSaS \mid \lambda$$

Gramer'in belirsiz olduğunu gösteriniz?



Soru

$$S \rightarrow aSB \mid bB$$

$$B \rightarrow aB \mid b$$

ifadesinde tüm B'leri silerek yeniden yazın?

$$S \rightarrow aSB$$

$$\rightarrow aSBB$$

$$\rightarrow aSBBB$$

$$\rightarrow aSBBBB$$

$$\rightarrow aSBBBBB$$

$$\rightarrow aSBBBBBB$$

$$S \rightarrow aSb \mid bb$$

Soru

$$S \rightarrow a \mid aA \mid c$$

$$A \rightarrow aB \mid \lambda$$

$$B \rightarrow Aa$$

Gereksiz türeyenleri silin?

0

Soru

$$S \rightarrow aSSS$$

$$S \rightarrow bb \mid \lambda$$

türeyimleri sil

$$S \rightarrow aSSS$$

$$aSSSaSSSaSSS$$

$$\downarrow \rightarrow aaaaa$$

$$S \rightarrow aSSS$$

$$S \rightarrow bb \mid \lambda$$

$$S \rightarrow aSSS$$

$$\downarrow \downarrow \downarrow$$

$$aaa$$

✓

Soru

$$S \rightarrow AaB \mid aaB$$

$$A \rightarrow \lambda$$

$$B \rightarrow bba \mid \lambda$$

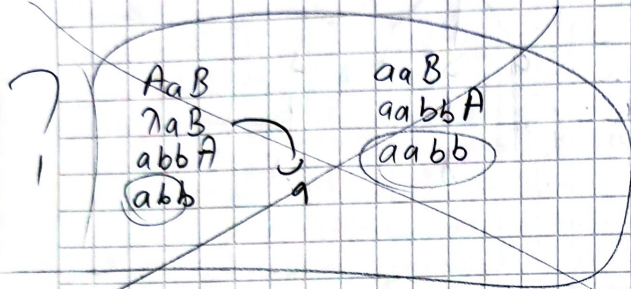
λ türetileri silin

$\rightarrow (a)$

$$S \rightarrow aaB \rightarrow aa$$

$$S \rightarrow aaB \rightarrow aabb$$

$$AaB \rightarrow abb$$



$$S \rightarrow a \mid aa \mid aabb \mid aabb$$

Chomsky

I. Aşama

$$S \rightarrow ABC$$

$$C \rightarrow a$$

$$A \rightarrow CCD$$

$$D \rightarrow b$$

$$B \rightarrow AE$$

$$E \rightarrow c$$

II. Aşama

$$S \rightarrow FC$$

$$F \rightarrow AB$$

$$C \rightarrow a$$

$$A \rightarrow GD$$

$$G \rightarrow CC$$

$$D \rightarrow b$$

$$B \rightarrow AE$$

$$E \rightarrow c$$

Greibach Form

$$A \rightarrow ax$$

$$a \in T, x \in V^*$$

$$S \rightarrow AB$$

$$A \rightarrow aA \mid bB \mid b$$

$$B \rightarrow b$$

Soru

$S \rightarrow aSS \mid a \mid b$ gramerini Chomsky - formuna dönüştürün.

Kelimeler

a

b

aab

abb

I. Aşama

$$S \rightarrow ASS \mid a \mid b$$

$$A \rightarrow a$$

II. Aşama

$$S \rightarrow BS \mid a \mid b$$

$$B \rightarrow AS$$

$$A \rightarrow a$$

Soru

$$S \rightarrow aSb \mid Sab \mid ab$$

$$① S \rightarrow ASB \mid SAB \mid AB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Chomsky formuna dönüştürün.

$$② S \rightarrow CB \mid SD \mid AB$$

$$C \rightarrow AS$$

$$D \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Soru

$$\begin{aligned} S &\rightarrow aSaaA \mid A \\ A &\rightarrow abA \mid bb \end{aligned}$$

Chomsky'e dönüştürün.

①

$$\begin{aligned} S &\rightarrow CS \mid CA \mid A \\ A &\rightarrow CBA \mid BB \end{aligned}$$

$$\begin{aligned} C &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

②

$$\begin{aligned} S &\rightarrow DE \mid A \mid BB \\ A &\rightarrow FA \mid BB \end{aligned}$$

$$\begin{aligned} D &\rightarrow CS \\ E &\rightarrow CC \\ F &\rightarrow CB \\ C &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

③

$$\begin{aligned} S &\rightarrow GA \mid FA \mid BB \\ A &\rightarrow FA \mid BB \end{aligned}$$

$$\begin{aligned} D &\rightarrow CS \\ E &\rightarrow CC \\ G &\rightarrow DE \\ F &\rightarrow CB \\ C &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Soru

$$S \rightarrow aSb \mid bSa \mid a \mid b \mid ab$$

Greibach formuna dönüştürün.

$$S \rightarrow aSb \mid \underbrace{bSA} \mid \underbrace{a \mid b \mid aB}$$

$$\begin{aligned} B &\rightarrow b \\ A &\rightarrow a \end{aligned}$$

$$\begin{aligned} A &\rightarrow ax \\ x &\in V^+ \end{aligned}$$

Soru

$$S \rightarrow aSb \mid ab \mid bb$$

Greibach formuna dönüştürün.

$$S \rightarrow aSB \mid aB \mid bB$$

$$B \rightarrow b$$

Soru

$$\begin{aligned} S &\rightarrow ABb \mid a \mid b \\ A &\rightarrow aaA \mid B \\ B &\rightarrow bAb \end{aligned}$$

Greibach formuna dönüştürün.

①

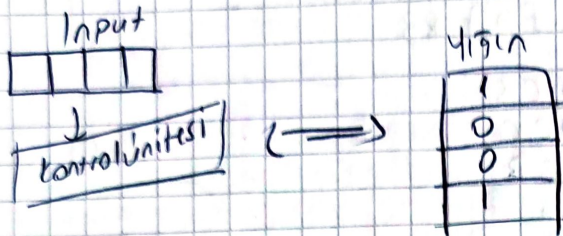
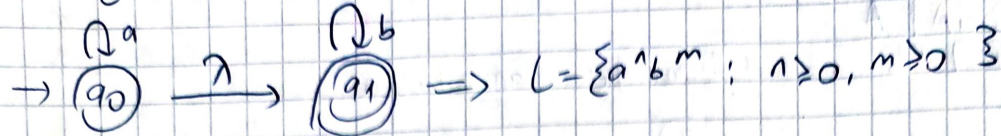
$$\begin{aligned} S &\rightarrow aaABb \mid a \mid b \\ A &\rightarrow aaA \mid bAb \\ B &\rightarrow bAb \end{aligned}$$

②

$$\begin{aligned} S &\rightarrow aCABDb \\ C &\rightarrow a \\ D &\rightarrow b \\ A &\rightarrow aCA \mid bAD \\ B &\rightarrow bAD \end{aligned}$$

Pushdown Automata

$$L = \{a^n b^n : n \geq 0\}$$



$$M = (Q, \Sigma, T, \delta, q_0, z, F)$$

$Q \rightarrow$ Sonlu durumlar

$\Sigma \rightarrow$ Giris Alfabeti

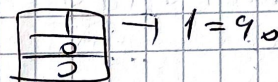
$T \rightarrow$ Yığın Alfabeti

$\delta \rightarrow$ Geçiş fonksiyonu

$$Q \times (\Sigma \cup \{\epsilon\}) \times T \rightarrow \text{Tanım}$$

$q_0 \rightarrow$ ilk durum

$z \rightarrow$ Yığın başlangıç sembolü



$F \rightarrow$ Sonlu durumlar

$$\delta(q_1, a, b) = \{(q_2, ab), (q_3, \lambda)\}$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{a, b\}$$

$$T = \{0, 1\}$$

$$z = 0$$

$$F = \{q_3\}$$

$$\delta(q_0, a, 0) = \{(q_1, 10), (q_3, \lambda)\}$$

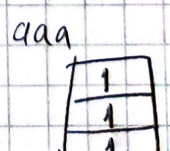
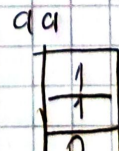
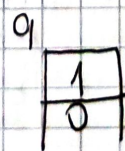
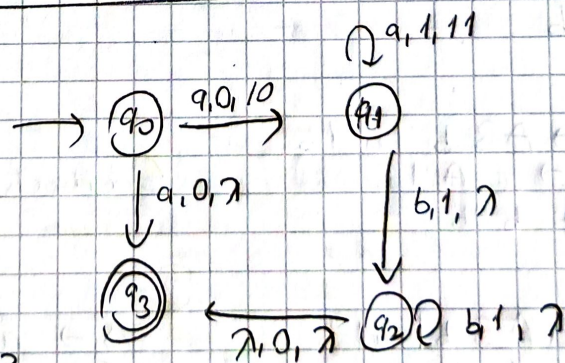
$$\delta(q_0, \lambda, 0) = \{(q_3, \lambda)\}$$

$$\delta(q_1, a, 1) = \{(q_1, 11)\}$$

$$\delta(q_1, b, 1) = \{(q_2, \lambda)\}$$

$$\delta(q_2, b, 1) = \{(q_2, \lambda)\}$$

$$\delta(q_2, \lambda, 0) = \{(q_3, \lambda)\}$$

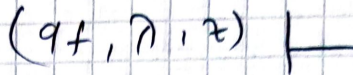


$$L = \{a^n b^n, n \geq 0\} \cup \{a\}$$

66

$$(90, 6, 07) \quad \perp$$

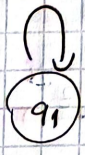
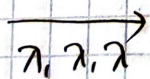
9.7.1



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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$$\times (90, 7, 02) \quad | -$$

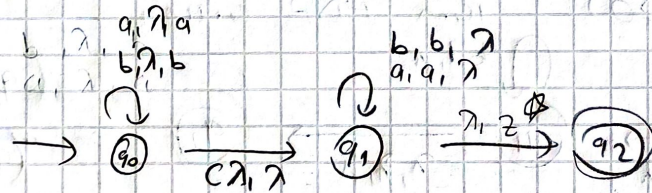
$\text{Na}, 117$

$$b, 1, \lambda$$

$$(90, \lambda, \lambda) \vdash$$

3

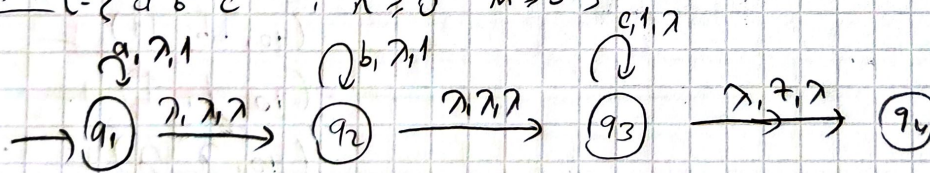
Soru $L = \{ w c w^R \mid w \in \{a,b\}^* \}$
 $\Sigma = \{a,b,c\}$ PDA tasarlayınız.

abcba
 abbcbbā

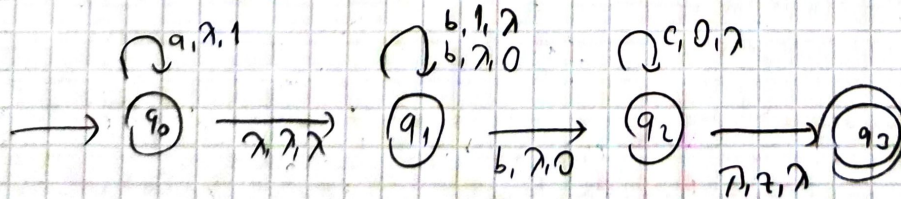


abbcbbā ✓
 bbcbba ✓
 bcbbā ✓
 cbba ✓
 bba ✓
 ba ✓
 a ✓
 λ ✓

Soru $L = \{ a^n b^m c^{n+m} \mid n \geq 0, m \geq 0 \}$



Soru $L = \{ a^n b^{n+m} c^m \mid n \geq 0, m \geq 1 \}$ için PDA tasarlayın.

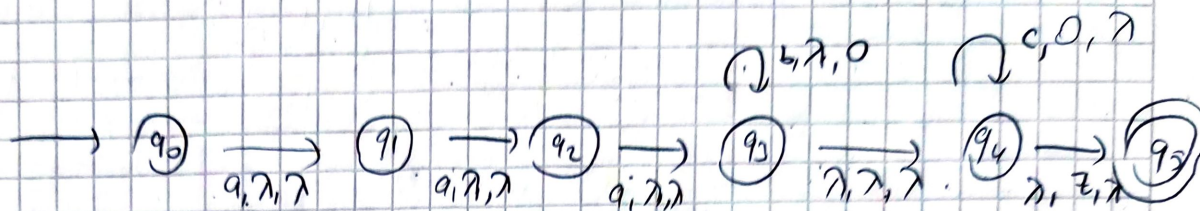


Soru

$$L = \{ a^3 b^n c^n ; n \geq 0 \}$$

PDA tasarlayın

aaa
aaabcc
aaabbbcc



Soru

$$M = (\{ q_0, q_1, q_2, q_3, q_4, q_5 \}, \{ a, b \}, \{ 0, 1, 2 \}, \delta, q_0, z, \{ q_5 \})$$

$$\delta(q_0, b, z) = \{ (q_1, 1, z) \}$$

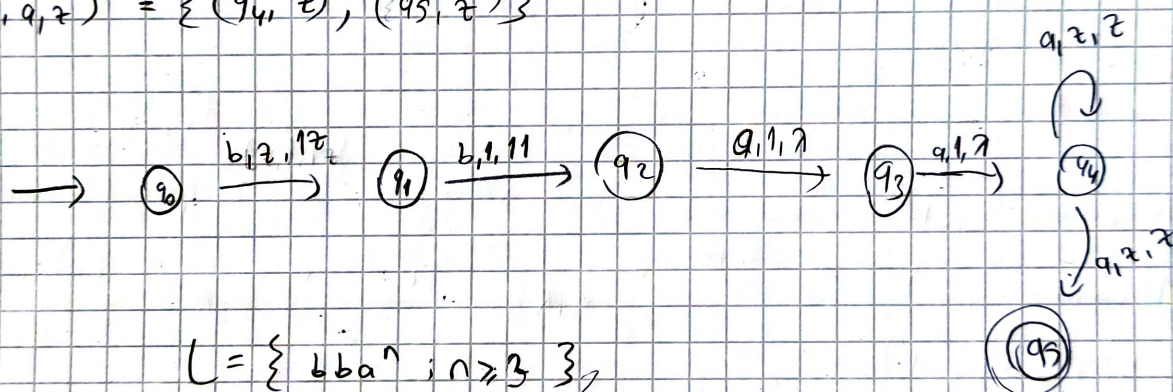
$$\delta(q_1, b, 1) = \{ (q_2, 11) \}$$

$$\delta(q_2, a, 1) = \{ (q_3, \lambda) \}$$

$$\delta(q_3, a, 1) = \{ (q_4, \lambda) \}$$

$$\delta(q_4, a, z) = \{ (q_4, z), (q_5, z) \}$$

dilini yazınız



$$L = \{ bba^n ; n \geq 3 \}$$

Soru

$$L = \{ a^n b^m ; n \leq m \leq 3n \}$$

PDA çiziniz

abbb
aa bbb
aa bbb b b

